Analog Electronics II: Op-amp & 555 Timer

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EEE 311

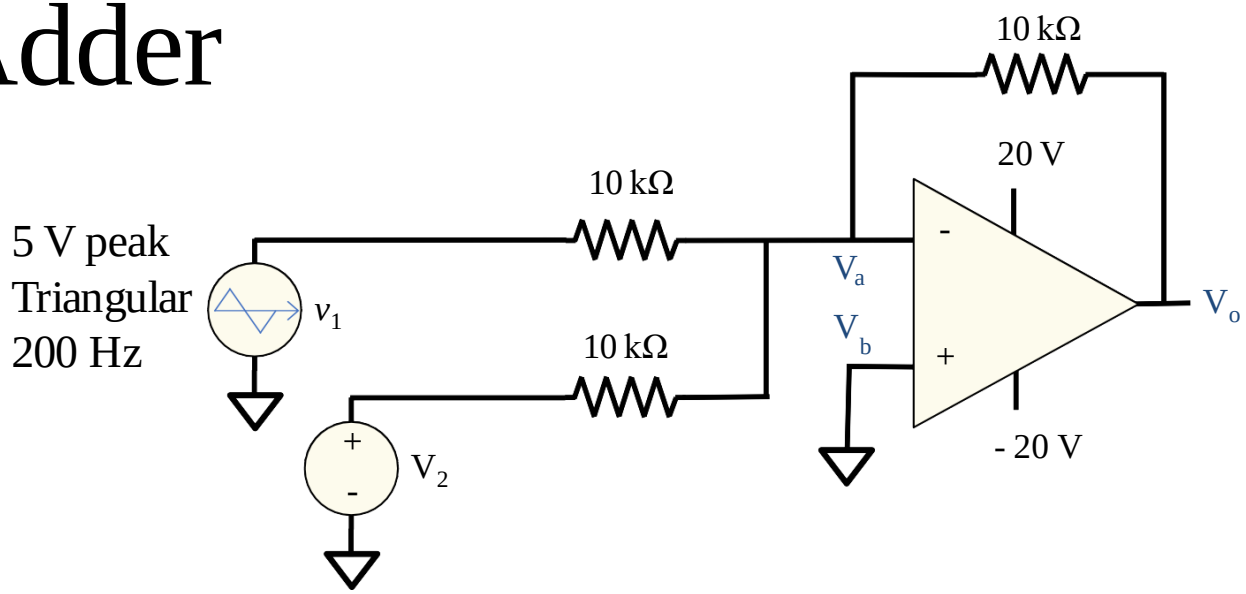
Lecture # 7

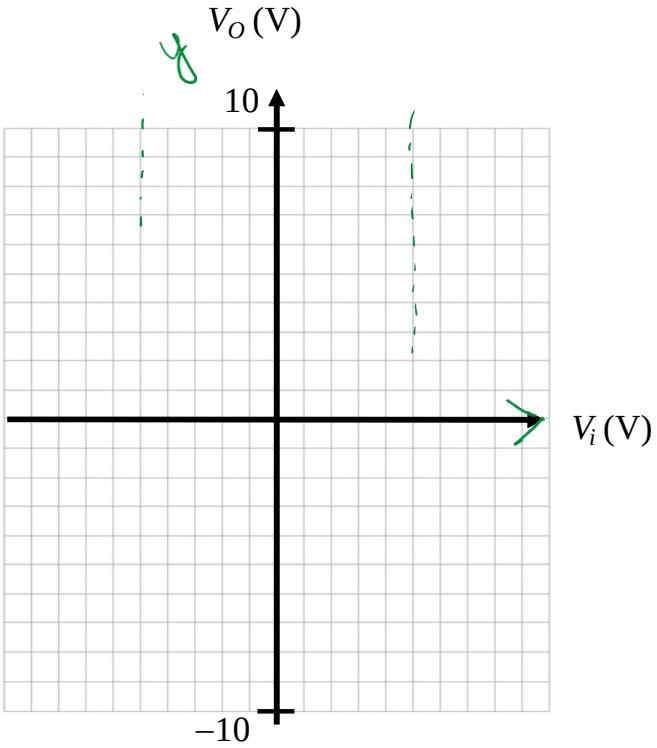
DC Offset Adder

Problem 21. Draw the output waveform and transfer characteristics for

i. $V_2 = +5$ V and

ii. $V_2 = -5$ V







Coughlin 3-2.3

Subtractor

$v_o = 5v_1 - 2v_2$ ⇒ we learn this with inv amp. & inv adder

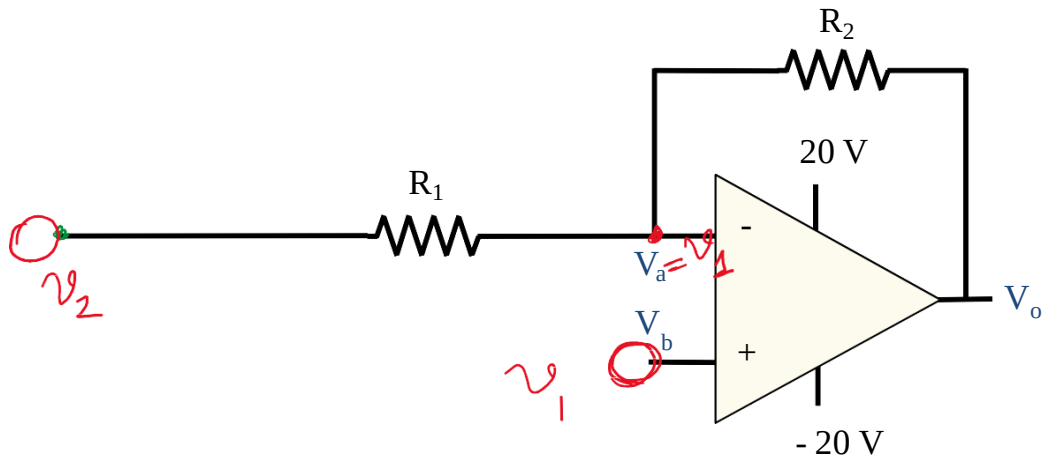
i) weighted subtrac^n:

ii) Difference amplifier:

$v_o = 5(v_1 - v_2)$

iii) Direct subtrac^n: (with unity gain)

$v_o = v_1 - v_2$



Mingle the inv amp. & non-inv amp. ?

Superpos^n:

v_1 active alone:

non-inv. amp.

$v_{o_1} = \left(1 + \frac{R_2}{R_1}\right) v_1$

* v_2 active alone

inv amp.

$v_{o_2} = -\frac{R_2}{R_1} \cdot v_2$

$v_o = v_{o_1} + v_{o_2}$

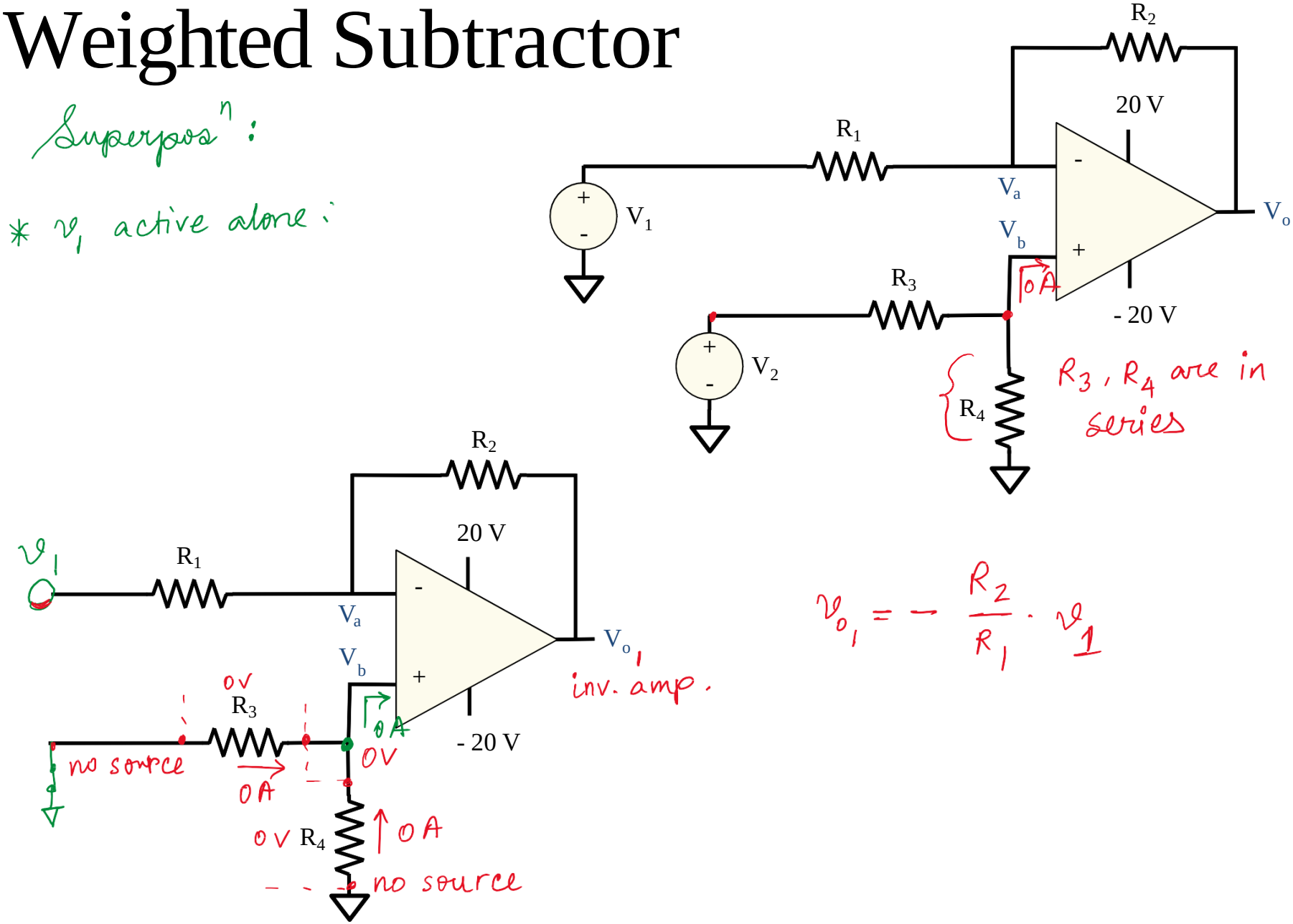
$= \left(1 + \frac{R_2}{R_1}\right) v_1 + \left(-\frac{R_2}{R_1}\right) v_2$

Source: Coughlin 3-2.3

Weighted Subtractor

Superposition:

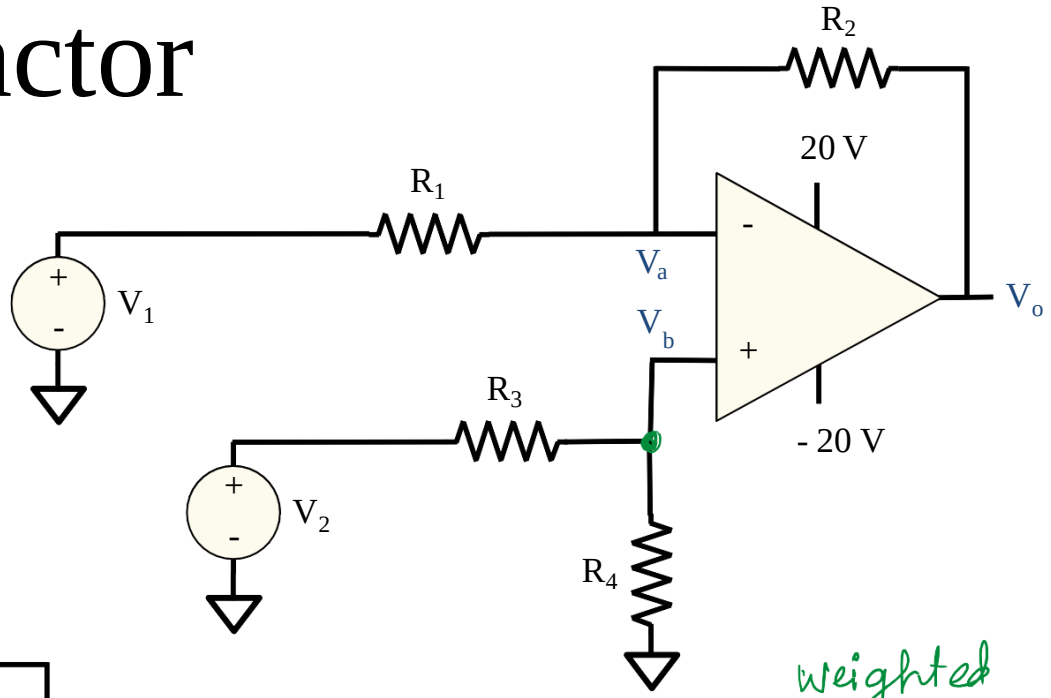
* v_1 active alone:



$$v_{o_1} = -\frac{R_2}{R_1} \cdot v_1$$

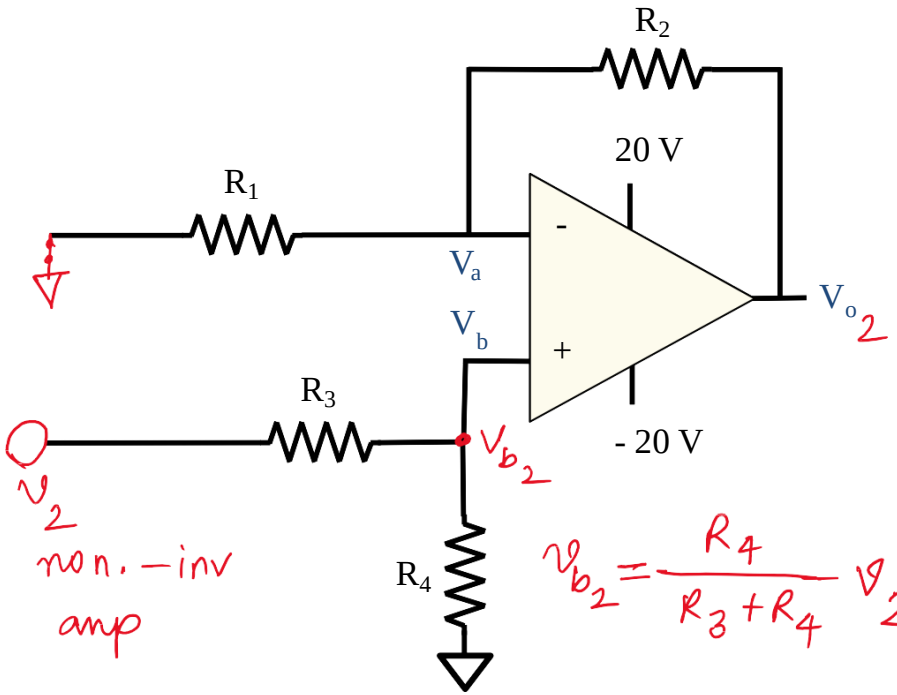
Source: Coughlin 3-2.3

Weighted Subtractor



$v_{o_2} = \left(1 + \frac{R_2}{R_1}\right) v_{b_2}$

$v_{o_2} = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right) v_2$



$v_{b_2} = \frac{R_4}{R_3 + R_4} v_2$

weighted subtractor

$v_o = v_{o_1} + v_{o_2}$

$= \underbrace{-\frac{R_2}{R_1}}_{A_{CL_1}} \cdot v_1 + \underbrace{\left(1 + \frac{R_2}{R_1}\right) \cdot \left(\frac{R_4}{R_3 + R_4}\right)}_{A_{CL_2}} v_2$

Source: Coughlin 3-2.3

Weighted Subtractor

design condⁿ

$R_4 = R_2$

$R_3 = R_1$

} diff. amp.

weighted subtractor ✓

$v_o = v_{o_1} + v_{o_2}$

$= \underbrace{-\frac{R_2}{R_1}}_{A_{cl_1}} \cdot v_1 + \underbrace{\left(1+\frac{R_2}{R_1}\right)\cdot\left(\frac{R_4}{R_3+R_4}\right)}_{A_{cl_2}} v_2$

$\Rightarrow \left(\frac{R_1+R_2}{R_1}\right)\left(\frac{R_4}{R_3+R_4}\right)\cdot v_2$ ($R_4 = R_2$)

$\Rightarrow \frac{R_1+R_2}{R_1}\cdot\frac{R_2}{R_1+R_2}\, v_2$

$v_o = -\frac{R_2}{R_1}\cdot v_1 + \frac{R_2}{R_1}\cdot v_2 = \frac{R_2}{R_1}(v_2 - v_1)$: Difference amp.

$v_o = v_2 - v_1$: Subtractor (unity gain) →

design condⁿ

$R_4 = R_2 = R_3 = R_1$

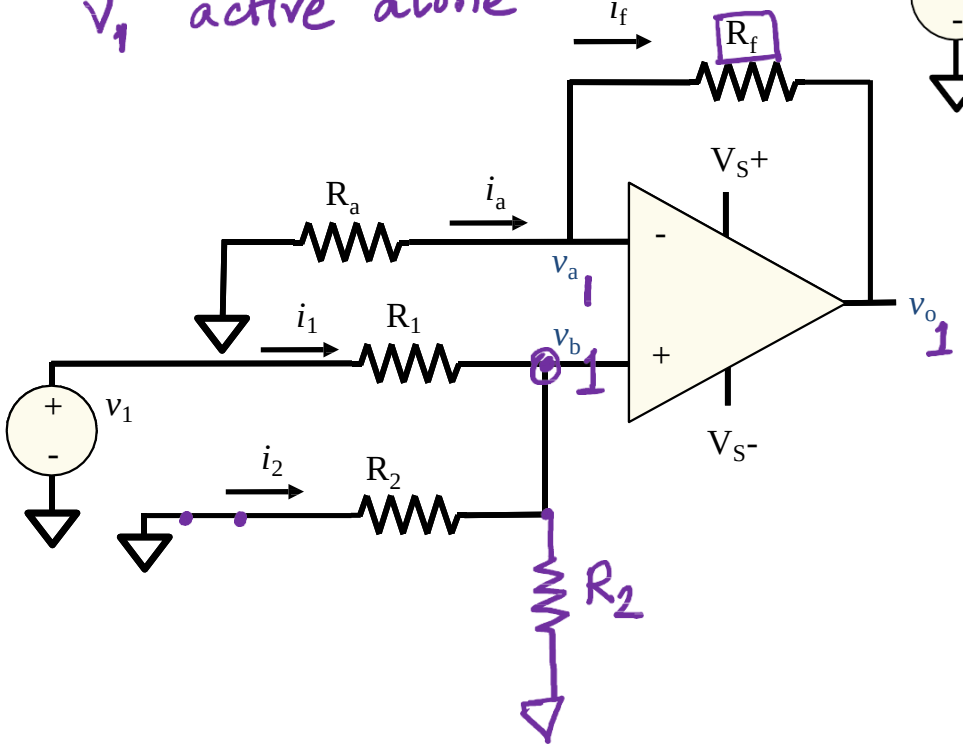
Source: Coughlin 3-2.3

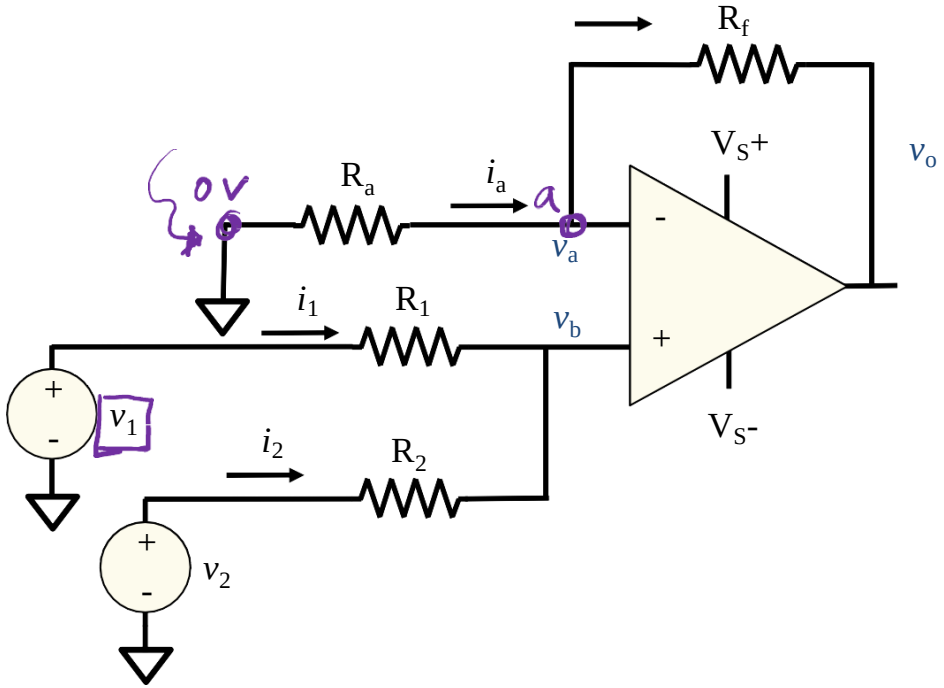
Non-inverting Adder

$$v_o = \left(1 + \frac{R_f}{R_a}\right) v_b \quad \cdots\cdots (1)$$

Superposition

V_1 active alone





$$v_{b_1} = \frac{R_2}{R_1 + R_2} \cdot v_1 \quad \cdots\cdots (11)$$

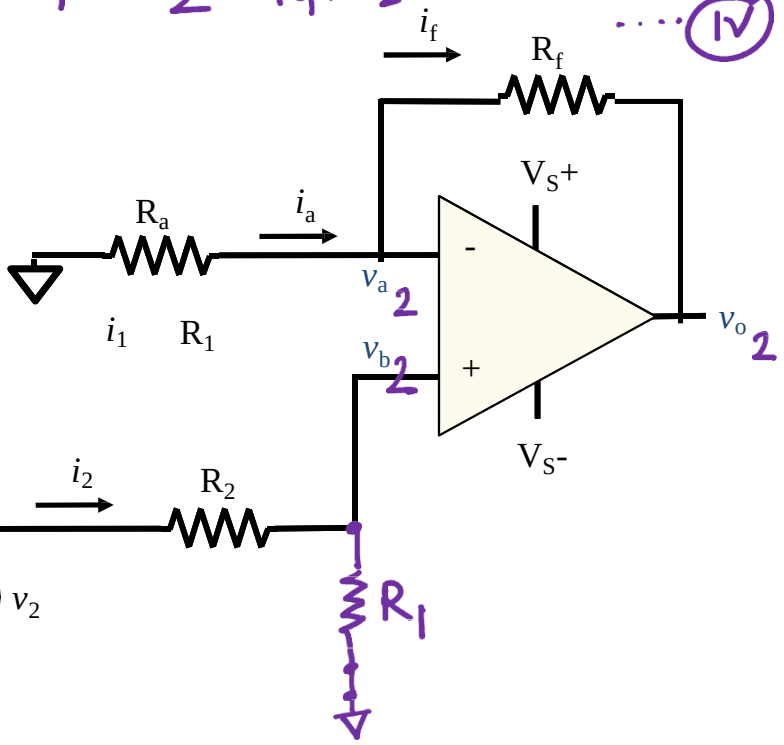
Non-inverting Adder

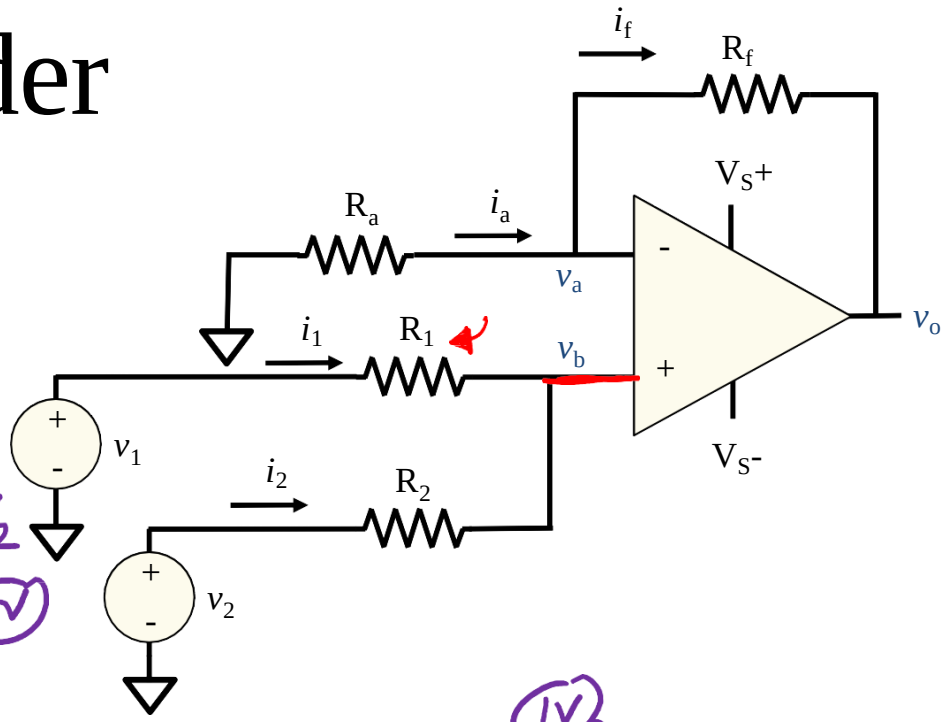
V_2 active alone

$$v_{b_2} = \frac{R_1}{R_1+R_2} \cdot v_2 \quad \cdots (III)$$

(II), (III) $\Rightarrow$

$$v_b = v_{b_1} + v_{b_2} = \frac{R_2}{R_1+R_2} \cdot v_1 + \frac{R_1}{R_1+R_2} \cdot v_2 \quad \cdots (IV)$$





(I) $\Rightarrow$ (IV)

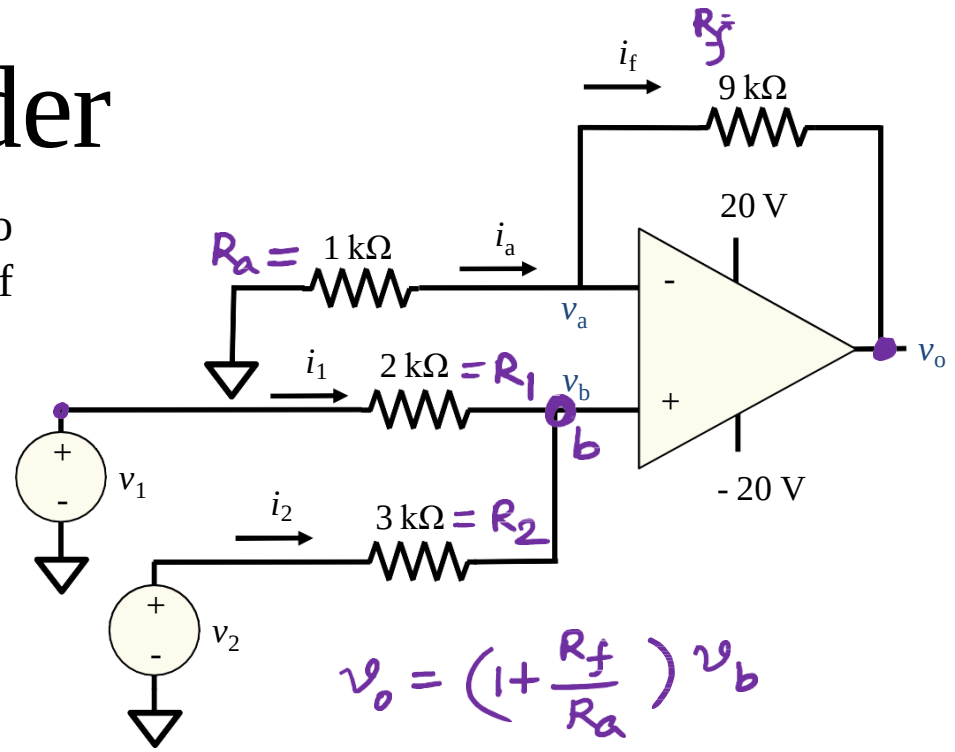
$$V_o = \left(1 + \frac{R_f}{R_a}\right)\left\{\frac{R_2}{R_1+R_2} V_1 + \frac{R_1}{R_1+R_2} \cdot V_2\right\}$$

$$= \left\{\left(1 + \frac{R_f}{R_a}\right)\left(\frac{R_2}{R_1+R_2}\right) V_1\right\} +$$

$$\left\{\left(1 + \frac{R_f}{R_a}\right)\left(\frac{R_1}{R_1+R_2}\right) V_2\right\}$$

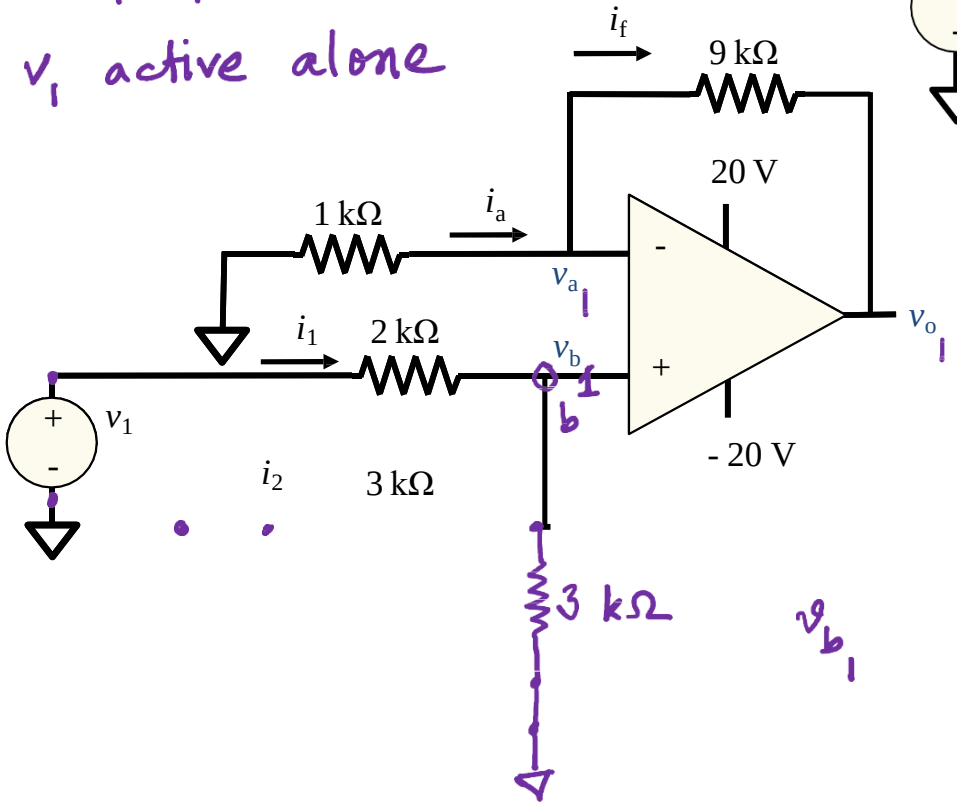
Non-inverting Adder

Problem 22. Use the superposition principle to find the output voltage of the circuit in terms of v_1 and v_2. Assume ideal conditions.



Superposition

v_1 active alone



$$v_o = \left(1 + \frac{R_f}{R_a}\right) v_b$$

$$= \left(1 + \frac{9\,k\Omega}{1\,k\Omega}\right) v_b$$

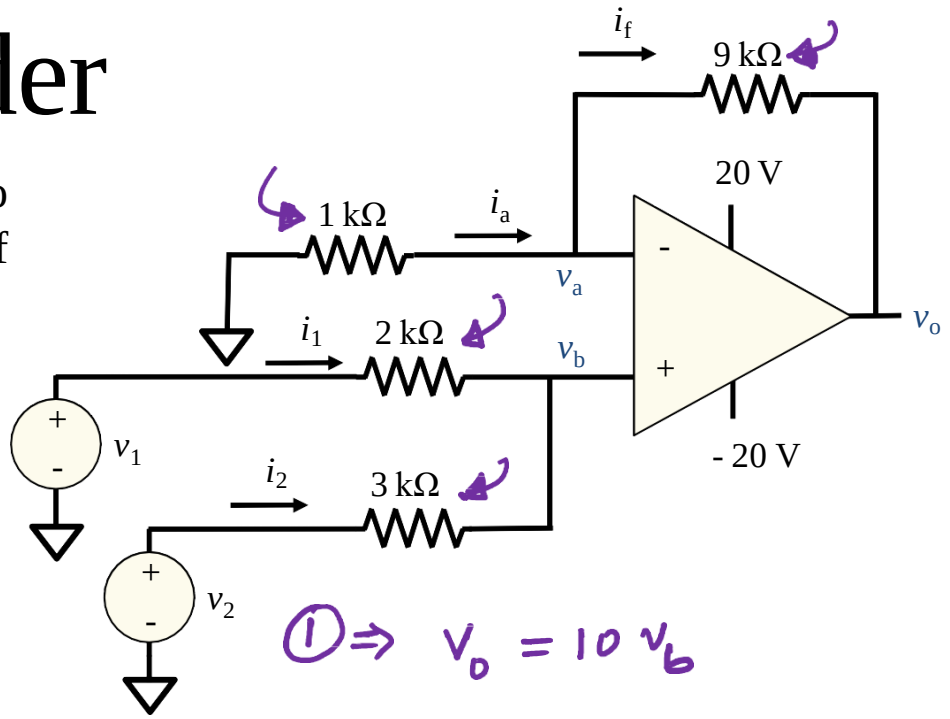
$$v_o = 10\, v_b \quad \cdots \quad (1)$$

$$v_{b_1} \quad \frac{3\,k\Omega}{(2+3)\,k\Omega} V_1 = \frac{3}{5} V_1 \quad \cdots \quad (11)$$

Ans. $v_o = 6v_1 + 4v_2$

Non-inverting Adder

Problem 22. Use the superposition principle to find the output voltage of the circuit in terms of v_1 and v_2. Assume ideal conditions.

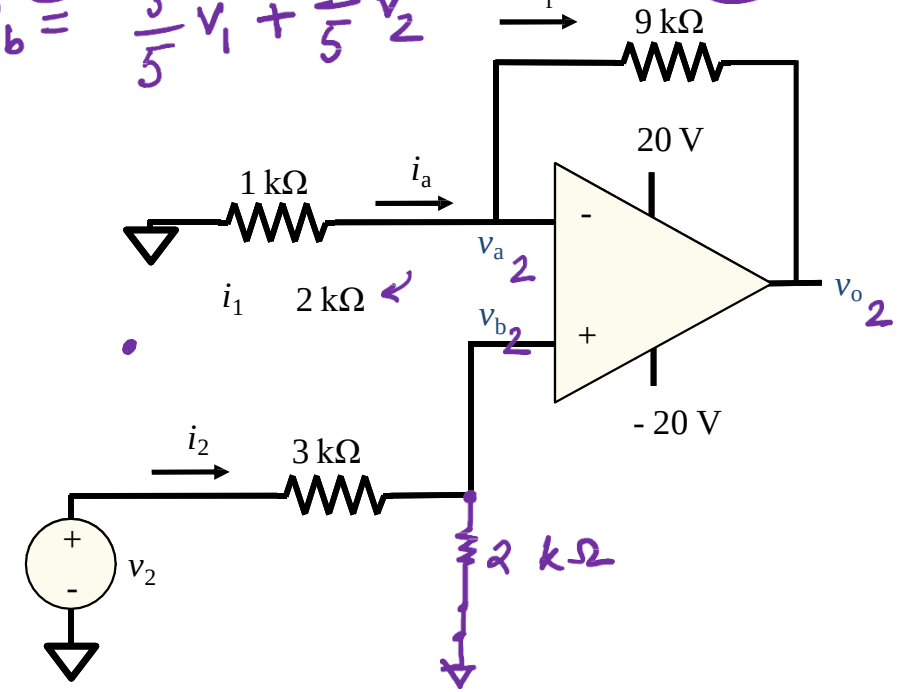


v_2 active alone

$$v_{b_2} = \frac{2\,k\Omega}{(2+3)\,k\Omega} v_2 = \frac{2}{5} v_2 \quad \cdots (III)$$

(II), (III) $\Rightarrow$

$$v_b = \frac{3}{5} v_1 + \frac{2}{5} v_2 \quad \cdots (IV)$$



(I) $\Rightarrow$ $V_o = 10\, v_b$

$$= 10\left(\frac{3}{5} v_1 + \frac{2}{5} v_2\right)$$

$$\Rightarrow V_o = 6 v_1 + 4 v_2$$

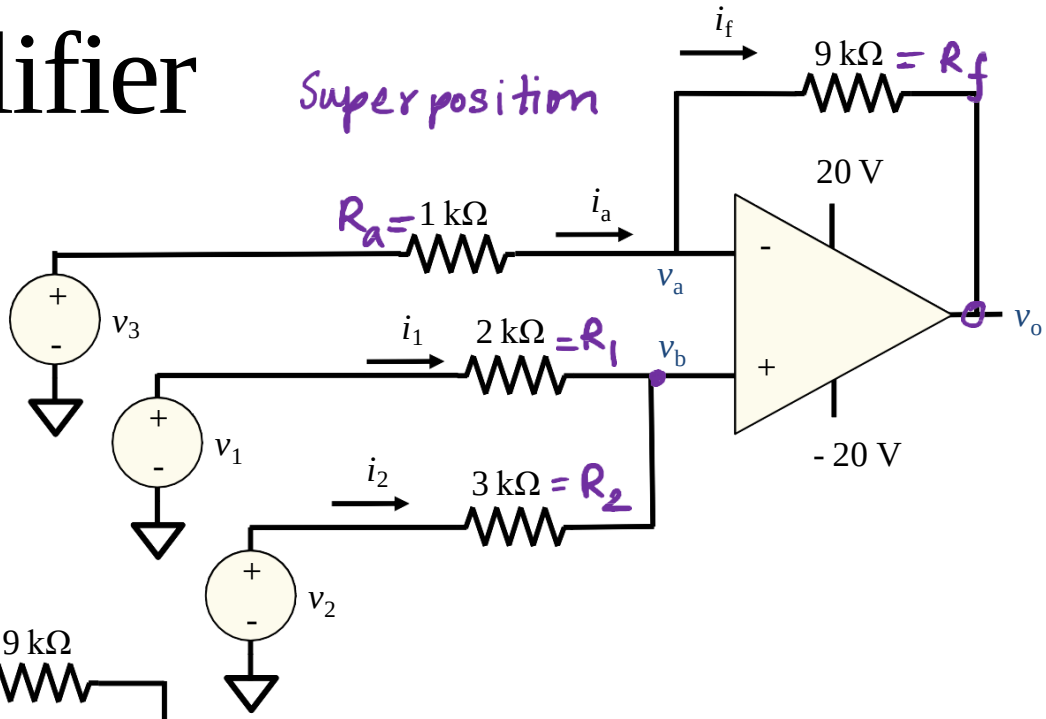
weighted adder

Ans. $v_o = 6v_1 + 4v_2$

Difference Amplifier

Superposition

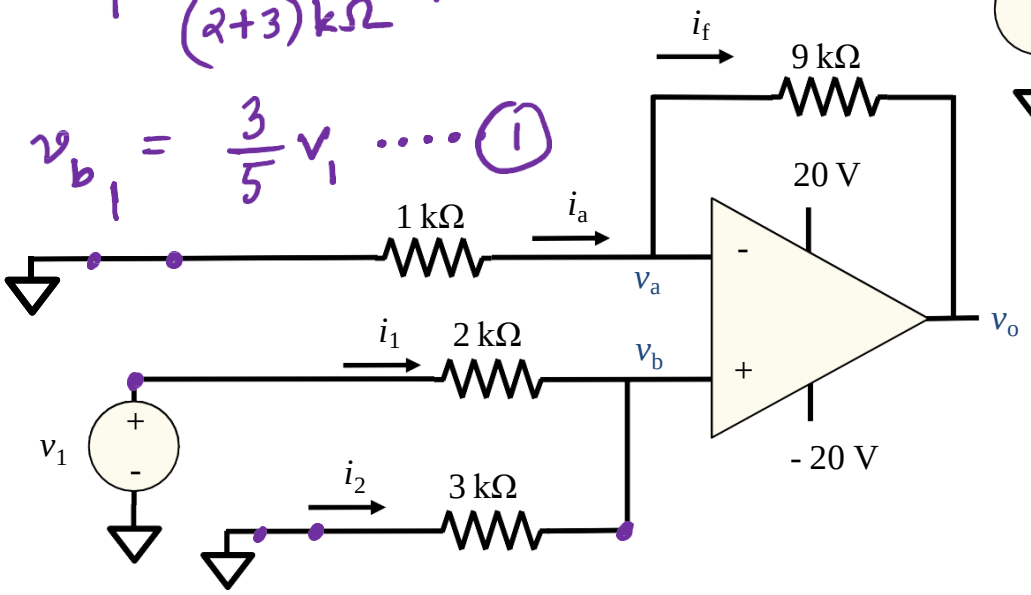
Problem 23. Use the superposition principle to find the output voltage of the circuit in terms of v_1, v_2, and v_3. Assume ideal conditions.



v_1 active alone

$$v_{b_1} = \frac{3\,k\Omega}{(2+3)\,k\Omega} \cdot v_1$$

$$v_{b_1} = \frac{3}{5} v_1 \quad \cdots (I)$$



$$v_{o_1} = \left(1 + \frac{9\,k\Omega}{1\,k\Omega}\right) v_{b_1}$$

$$= 10 \cdot \frac{3}{5} v_1$$

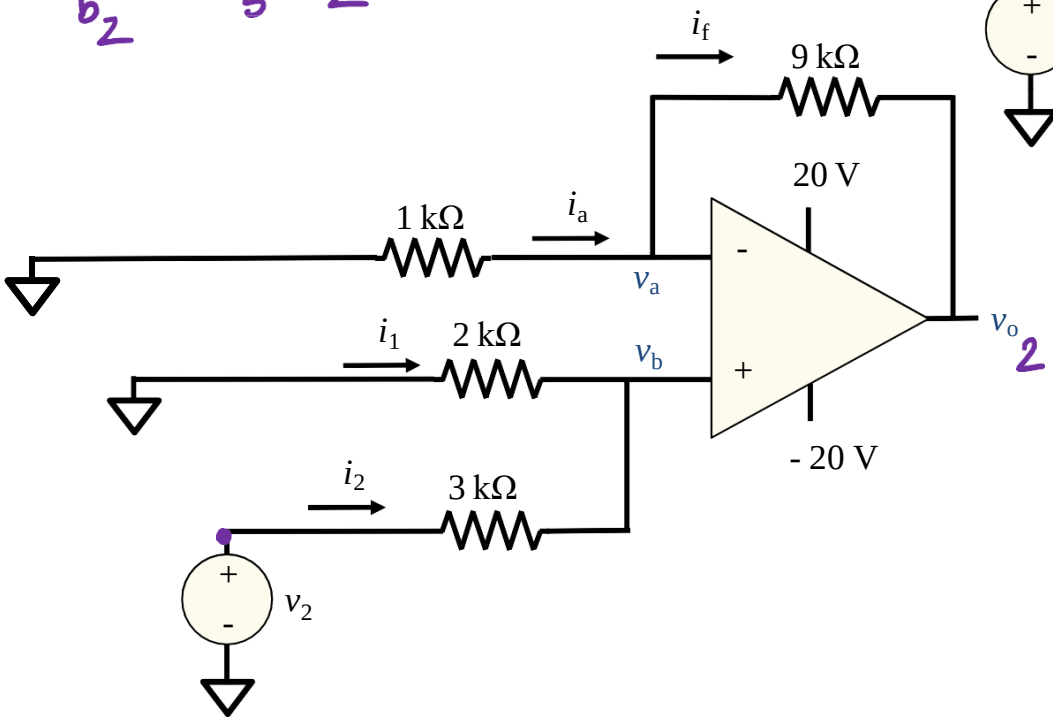
$$v_{o_1} = v_1 \quad \cdots (II)$$

Difference Amplifier

V_2 active alone

$$v_{b_2} = \frac{2\,k\Omega}{(2+3)\,k\Omega} \cdot V_2$$

$$v_{b_2} = \frac{2}{5} V_2 \quad \cdots \text{(III)}$$



$$v_{o_2} = \left(1 + \frac{9\,k\Omega}{1\,k\Omega}\right) v_{b_2}$$

$$= 10 \cdot \frac{2}{5} V_2$$

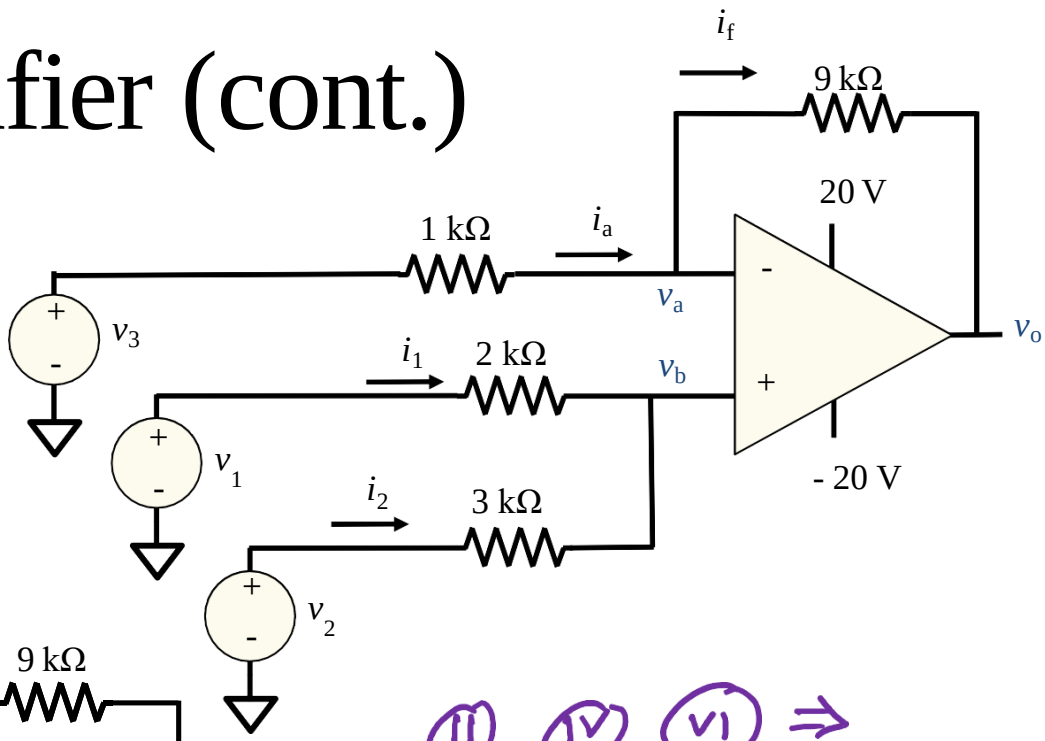
$$v_{o_2} = 4 V_2 \quad \cdots \text{(IV)}$$

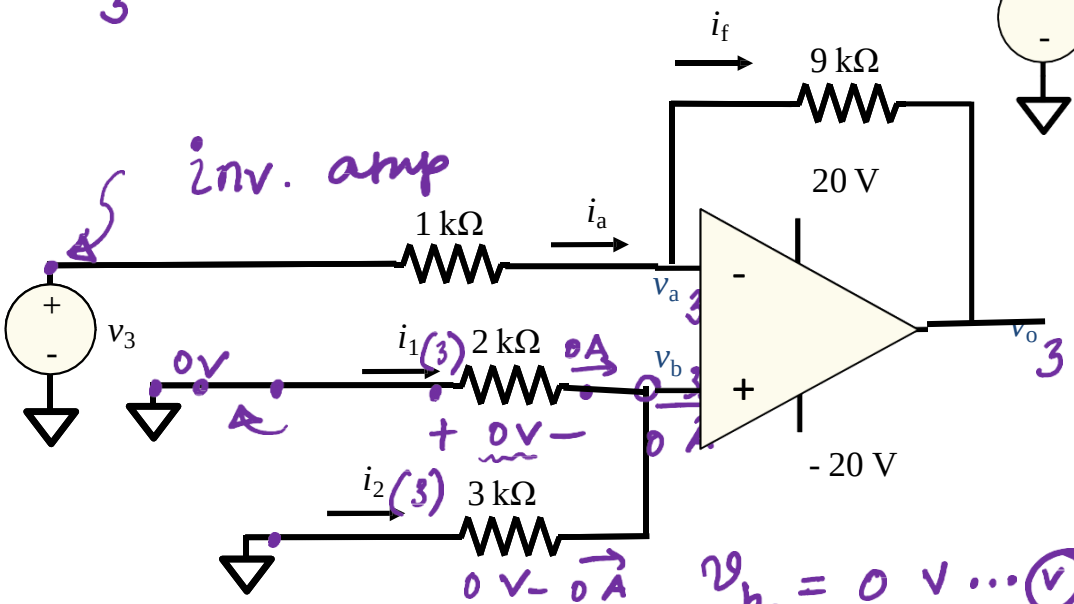
Difference Amplifier (cont.)

V_3 active alone

$$v_{o_3} = -\frac{9\,k\Omega}{1\,k\Omega} \cdot v_3$$

$$v_{o_3} = -9 v_3 \quad \cdots (VI)$$





$v_{b_3} = 0\ V \cdots (V)$

(II), (IV), (VI) ⇒

$$v_o = 6v_1 + 4v_2 - 9v_3$$

Ans. $v_o = 6v_1 + 4v_2 - 9v_3$